

복사전달 특강

Special Topics in Radiative Transfer

Lecture 8
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Based on: Chapters 5 & 6 of Noebauer & Sim (Living Reviews in Computational Astrophysics)

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Chapter 5: Monte Carlo Quanta

Overview of discretization paradigms

What are MC Quanta?

- MCRT discretizes radiation into a large number of packets.
- Each packet carries: position, propagation direction, frequency, energy/weight, and (optionally) a Stokes polarization vector.

5.1 Photon Packet Scheme

- Quanta = bundles of physical photons
- Weights encode photon count per packet
- Non-uniform & variable weights → variance reduction
- Dominant in dust RT simulations

5.2 Energy Packet Scheme

- Quanta = parcels of radiant energy
- Packet energy = its weight (conserved)
- Indivisible: no splitting during propagation
- Developed by L. Lucy (1999)

Key difference: photon packets conserve photon number; energy packets conserve energy.

5.1 Photon Packet Scheme

Historically dominant approach in astrophysical MCRT

Historical Inspiration

- Inspired by nature's own discretization.
- Early MCRT studies (Auer 1968; Avery & House 1968) referred to quanta simply as 'photons'.

Monte Carlo photon packets

- Each MC packet represents a vast number of real photons.
- The simulation count is completely insignificant compared to the actual radiation field.

Packet Weights

- Weights encode the physical photon count per packet.
- In this scheme, the MC quanta are typically referred to as ***photon packets*** or simply ***packets***. Non-uniform and variable weights reduce variance (MC noise) dramatically — a form of biasing (Sect. 9.4).

Applications

- Widely used in dust RT (Whitney 2011; Steinacker et al. 2013).
- Most astrophysical MCRT codes rely on this scheme with sophisticated weight manipulations.

5.2 Energy Packet Scheme

Lucy's indivisible energy packet formalism

Core Principle

- Packets are parcels of radiant energy.
- The packet's energy serves as its weight and is strictly conserved — packets are **indivisible** and **indestructible** (except at domain boundaries).

Indivisibility

- No splitting.
- Multiple outcome channels handled by probabilistic branching, preserving total energy throughout.

Energy Conservation

- Strict local energy conservation by construction in RE problems.
- Frequency can change at each interaction; CMF energy stays fixed.

Photon Number

- NOT conserved (physically correct: e.g. recombination cascades don't conserve photon number).
- A packet may represent varying photon counts during its lifetime.
- Many physical radiation-matter processes (e.g. recombination cascades or fluorescence) do not conserve the number of photons.

Key Advantages

- Effort proportional to energy carried, not photon count
- Avoids accumulation of negligible-weight quanta (no elimination scheme needed)
- Widely used: stellar winds, SN ejecta (Lucy 2005; Kerzendorf & Sim 2014)

Compton Scattering: Two Schemes Compared

A worked example illustrating the philosophical difference

Process: $e_i^- + \gamma_i \rightarrow e_f^- + \gamma_f$ (photon transfers recoil energy to electron; $\nu_f < \nu_i$)

Photon Packet Approach

- Packet scatters $\rightarrow$ photon number fixed
- Frequency reduced ($\nu_f < \nu_i$)
- Packet energy decreases accordingly

✓ Good for: computing Comptonized spectra
✗ Risk: many low-energy packets accumulate, each costing the same compute per scatter

Energy Packet Approach

- Track energy flow, not photon count
 - $F_\gamma = E_f^\gamma / E_i^\gamma$ of the incident photon energy $\rightarrow$ is passed to the outgoing photon packet.
 - $F_e = E_f^e / E_i^\gamma = 1 - F_\gamma \rightarrow$ goes to the electron kinetic pool (k-packet)
- ✓ Energy strictly conserved
✓ Effort $\propto$ energy carried (ideal for heating applications)

Which is better?

- Neither is universally superior. Photon packets suit spectral calculations (Pozdnyakov et al. 1983).
- Energy packets suit heating applications (SN ejecta powered by radioactive decay, Lucy 2005).

5.3 Packet Initialization

Assigning initial properties via Monte Carlo sampling

Each packet is initialized with:

Position · Propagation direction · Frequency · Energy / Weight · Stokes vector (if polarization included)

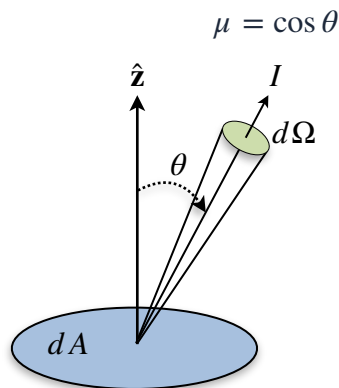
Case A: Uniform, Thermal Radiation from a volume ΔV

- Each of N packets carries: $\varepsilon = \Delta V \cdot a_R T^4 / N$
($a_R = (4/c)\sigma_{\text{SB}}$)
- Position: uniform random in cuboid ΔV
 $x = x_0 + \xi_1(x_1 - x_0)$
 $y = y_0 + \xi_2(y_1 - y_0)$
 $z = z_0 + \xi_3(z_1 - z_0)$
- Direction: isotropic (Eq. 26-27): $\mu = 2\xi_1 - 1$ and $\phi = 2\pi\xi_2$
- Frequency: sampled from Planck function

Case B: Photospheric Boundary

- $N = 4\pi R_{\text{phot}}^2 \sigma_{\text{SB}} T_{\text{phot}}^4 \Delta t / \varepsilon$
- Position: $r = R_{\text{phot}}$ (on the photosphere)
- Direction: $\mu = \sqrt{\xi_1}$ and $\phi = 2\pi\xi_2$
(flux-based; limb darkening neglected; isotropic radiation)
- Frequency: sampled from Planck function

- Initialization is performed in the co-moving frame (CMF).
- Lab-frame (LF) properties are obtained via frame transformations (see Sect. 8 for expanding media).



Contribution of I to the flux:

$$P(\mu, \phi) d\mu d\phi dA = I \cdot d\mu d\phi (dA \cdot \mu)$$

for $0 \leq \mu \leq 1$ ($0^\circ \leq \theta \leq 90^\circ$)

$$\therefore P(\mu, \phi) = \mu \frac{1}{2\pi} \text{ after normalization}$$

$$\int_0^\mu d\mu' \mu' = \xi_1 \quad \text{and} \quad \int_0^\phi \frac{1}{2\pi} d\phi' = \xi_2$$

Chapter 6: Propagation of Quanta

The core computational loop of any MCRT simulation

- After initialization, packets are propagated through the domain.
- This constitutes the bulk of MCRT runtime.

1. Draw τ

Sample optical depth:

$$\tau = -\ln \xi$$

2. Locate interaction

$$\text{Invert } \int_0^l dl' \chi(l') = \tau$$

to find distance l .

Here, $\chi(l')$ is the opacity.

3. Compare distances

distance l vs. cell boundary,
vs. time step end

→ take minimum,
assign the shortest distance to l .

4. Move packet

- Advance by l
- Update time stamp $t = t + l/c$

5. Perform event

- Scattering
- absorption, or
- cell crossing

6. Termination?

- Escape through one of the domain boundaries
- Absorption interaction
- Reaching the end of a pre-defined time interval.

Fig. 2: Packet Propagation Algorithm

The two-loop structure of the MCRT propagation algorithm

Outer Loop: Packet Loop

- Iterates over all MC packets in the population
- Each packet processed through propagation loop
- Returns to outer loop once packet terminates

Inner Loop: Propagation Loop

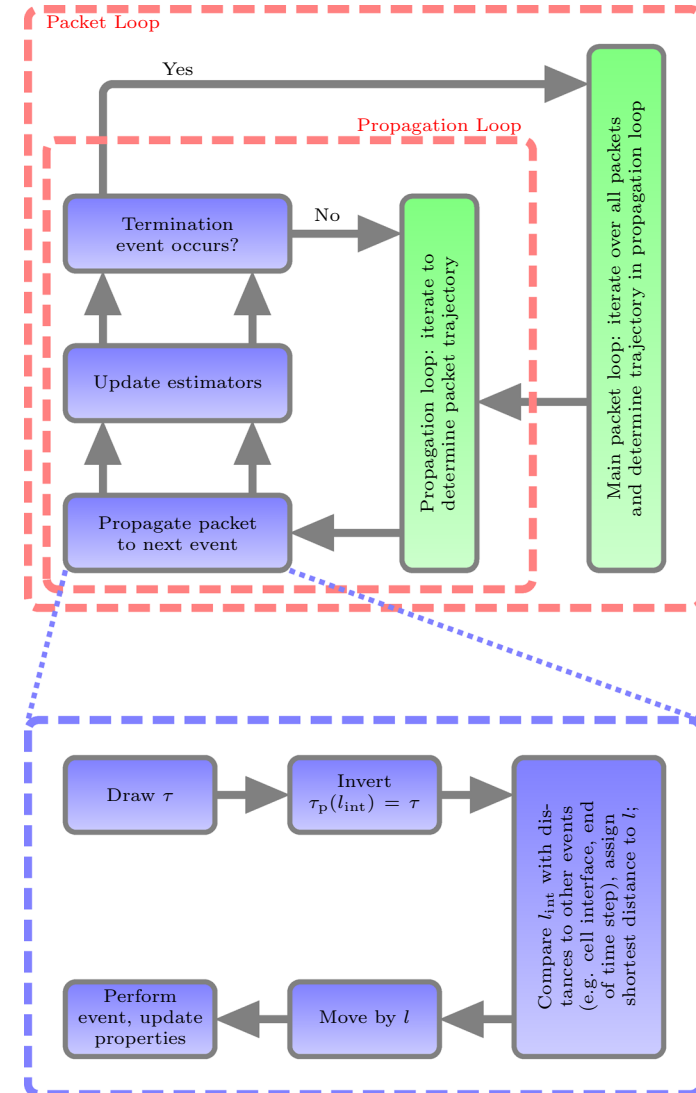
- Moves packet step by step through domain
- Each step: draw $\tau \rightarrow$ invert $\rightarrow$ move $\rightarrow$ event
- Loop continues until termination condition

Update Estimators

- At each step, packet properties are recorded to reconstruct radiation field quantities (energy density, flux, etc.) from the ensemble. See Sect. 9.

Illustration of the packet process.

- At its core, the main processing loop works through the entire packet population, propagating each packet individually. In this loop, each packet is moved through the domain until a termination event occurs. Depending on the specific MCRT application, this may for example be an absorption interaction, escaping through the domain boundaries or reaching the end of a pre-defined time interval.
- The instruction “update estimators” refers to all activities related to recording packet properties that are used to reconstruct physical information about the radiation field from the packet ensemble.



6.2 Absorption: Discrete vs. Continuous

Two complementary approaches to handling absorptive opacity

Discrete Interaction Events

- Packet propagates until τ threshold reached
- Absorption terminates packet at that point
- In RE (Lucy 1999a): immediately re-emitted
- Required for line interactions where absorption $\leftrightarrow$ re-emission must be linked
- Used in: Abbott & Lucy 1985; Long & Knigge 2002

Continuous Weight Degradation

- Weight decays along flight path:

$$w(l) = w_0 \cdot \exp(-\tau_a), \text{ where } \tau_a(l) = \int_0^l dl' \chi_a(l') \text{ is the optical}$$

depth due to the absorption opacity χ_a .

- Smoother $\rightarrow$ lower MC noise for small opacity
- Disconnects absorption from re-emission; new packets must be injected separately
- Suits pure-absorption problems
- Hybrid schemes common (continuum + lines)

This review focuses primarily on the discrete interaction approach; most principles also apply when weight degradation is used for continuum opacity.

6.3 Numerical Discretization & Material Properties

Implementing MCRT on a computational grid

- Real applications require a spatial grid dividing the domain into cells. Material properties (density, temperature, velocity) are represented discretely per cell.

Material State per Cell

- Sets local opacity $\rightarrow$ rate of optical depth accumulation [Eq. 44].
- Also dictates re-emission characteristics in scattering events.

Cell Crossing Events

- When a packet crosses a cell boundary, opacities are recalculated.
- τ may be redrawn (exploiting infinite divisibility of the exponential distribution).

Opacity Integration

- Within a constant-opacity cell: $l = \tau/\kappa$ (trivial inversion of Eq. 44).
- More complex for fast flows where opacity is frequency/direction dependent (Sect. 8).

Grid Types

- Cartesian, spherical, cylindrical, or unstructured — chosen based on problem geometry.
- Interpolation between cells is possible but adds computational cost.

6.4 Absorption & Scattering Interactions

Handling multiple opacity sources probabilistically

- Total opacity: $\kappa_{\text{tot}} = \kappa_s + \kappa_a$
- Interaction distance: $l = \tau / \kappa_{\text{tot}}$

Decision at the interaction point:

Draw $\xi \in [0, 1)$

If $\xi \leq \kappa_s / (\kappa_s + \kappa_a) \rightarrow \text{SCATTER}$

Else $\rightarrow \text{ABSORB}$ (remove packet, or re-emit in RE applications)

After Scattering (isotropic coherent example):

- New direction drawn: $\mu = 2\xi_1 - 1$, $\phi = 2\pi\xi_2$
- Frequency unchanged (coherent); new τ drawn; accumulated optical depth reset
- Multiple scatterings handled naturally — a major strength of MCRT over deterministic methods!
- Easily extended: anisotropic scattering simply changes the directional sampling rule

6.5 Propagation Example: Homogeneous Sphere

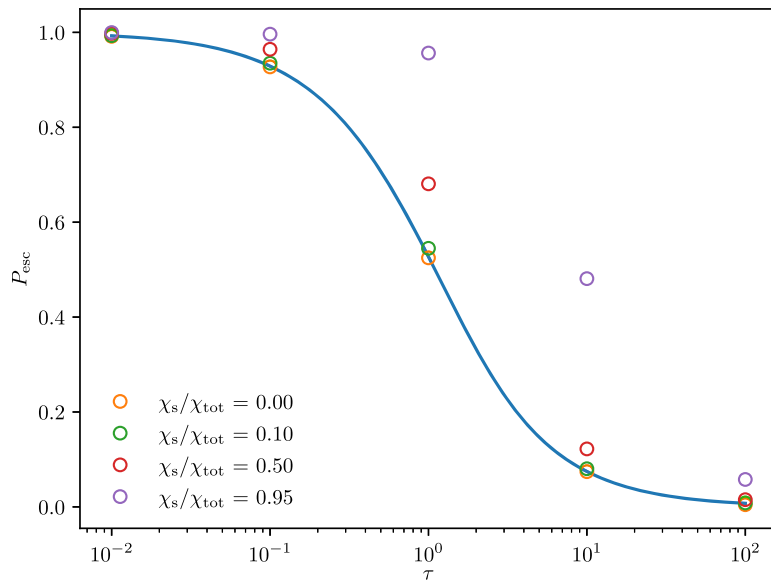
MCRT test case — escape probability from a uniform sphere

- Test problem: photon escape from a sphere of uniform density and emissivity.
- Analytic solution available for pure absorption (see Appendix A.1).
- Packets initialized at $r = R\xi^{1/3}$, direction $\mu = 2\xi - 1$.

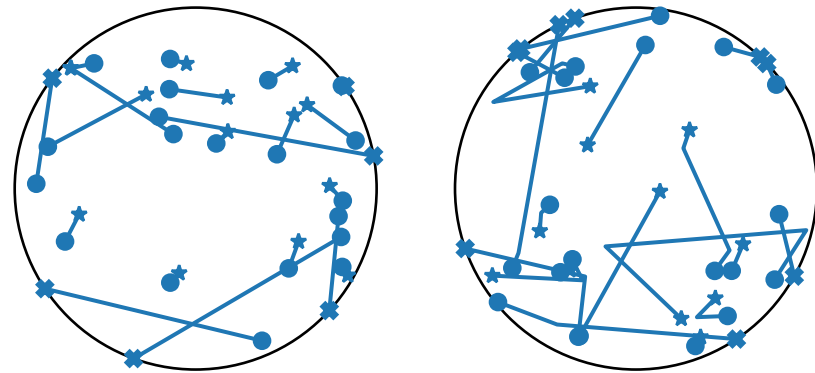
Python Code

Escape probability from a uniform sphere

<https://github.com/unoebauer/mcrtreview-tools>



Escape probability P_{esc} vs. total optical depth τ for different scattering fractions. Solid line = analytic solution (pure absorption). MCRT agrees well.



Sample packet trajectories for $\tau = 2$.

Left: $\kappa_s = 0$ (pure absorption).

Right: $\kappa_s = \kappa_a$. $\bullet$ = start, $\star$ = absorbed, $\times$ = escaped.

6.6 Time-Dependent MCRT

Extending steady-state MCRT to evolving radiation fields

- Time-dependence requires minimal changes: each packet carries an internal clock t , updated as $\Delta t = l/c$ after each propagation step.

Time Stamp

- Each packet: t initialized from emission process.
- Updated by $\Delta t = l/c$.

Time Grid

- Simulation subdivided into time steps t_n .
- Material state constant per step.

Interruption Rule

- Packet reaching t_{n+1} is paused; state saved.
- Resumes in next cycle.

Initialization in time-dependent problems:

- Initial radiation field packets: t = starting time of current simulation phase
- Source packets (continuous emitter): t drawn uniformly over current time step duration
- Material state updated between time steps (coupling with hydrodynamics → Sect. 11)
- Problems arising from very rapid material state changes: addressed in Sect. 10

Team Project

Thompson scattering by an electron gas at temperature T

The probability density function for the velocity component u of a scattering electron parallel to the incoming photon is

$$P(v) = (\text{cross section} = \text{constant}) \cdot \left(\frac{m_e}{2\pi k_B T} \right)^{1/2} \exp \left(-\frac{m_e v^2}{2k_B T} \right) = \frac{1}{\sqrt{\pi v_{\text{th}}^2}} \exp \left(-\frac{v^2}{v_{\text{th}}^2} \right)$$

$$\text{where } v_{\text{th}} = \sqrt{\frac{m_e}{2k_B T}}$$

- (1) Initial photon location: $(x, y, z) = (0, 0, 0)$.
- (2) Choose an initial photon direction vector ($\mathbf{\Omega}_i$).
- (3) Choose an optical depth, where the photon is scattered: $\tau = -\ln \xi \leftarrow \xi \in U(0, 1)$
- (4) Choose a velocity component v_z parallel to the incoming photon from the above PDF.
- (5) Choose two velocity components v_x and v_y perpendicular to the photon direction.

(6) Transform the frequency from LF to CMF: $\nu = \nu_i - \frac{v_z}{c} \nu_i$

(7) Choose scattering angles μ and ϕ in the CMF.

(8) Update the photon propagation direction: $\mathbf{\Omega}_i \rightarrow \mathbf{\Omega}_f$ using μ and ϕ

(9) Transform the frequency from the CMF to LF: $\nu_f = \nu + \frac{\mathbf{v} \cdot \mathbf{\Omega}_f}{c} \nu$

(10) Goto step (3) until the photon escape the sphere.

Choose three normal Gaussian random numbers:

$$\xi_1, \xi_2, \xi_3 \sim \text{Gauss}(0, 1).$$

Then,

$$v_z \sim v_{\text{th}} / \sqrt{2} \cdot \xi_1$$

$$v_x \sim v_{\text{th}} / \sqrt{2} \cdot \xi_2$$

$$v_y \sim v_{\text{th}} / \sqrt{2} \cdot \xi_3$$

Summary

Ch. 5: Two Discretizations

Photon packets conserve photon number; energy packets conserve energy. Choice depends on scientific goal — neither universally superior.

Ch. 5: Initialization

Packets draw position, direction, and frequency from appropriate distributions (Planck for frequency; isotropic or flux-based for direction).

Ch. 6: Propagation Loop

Draw $\tau = -\ln \xi \rightarrow$ locate interaction $\rightarrow$ move to minimum-distance event $\rightarrow$ scatter or absorb $\rightarrow$ repeat until termination condition.

Ch. 6: Scattering Strength

Multiple, anisotropic scattering is handled naturally in MCRT — a major advantage over deterministic RT solvers with non-local coupling.